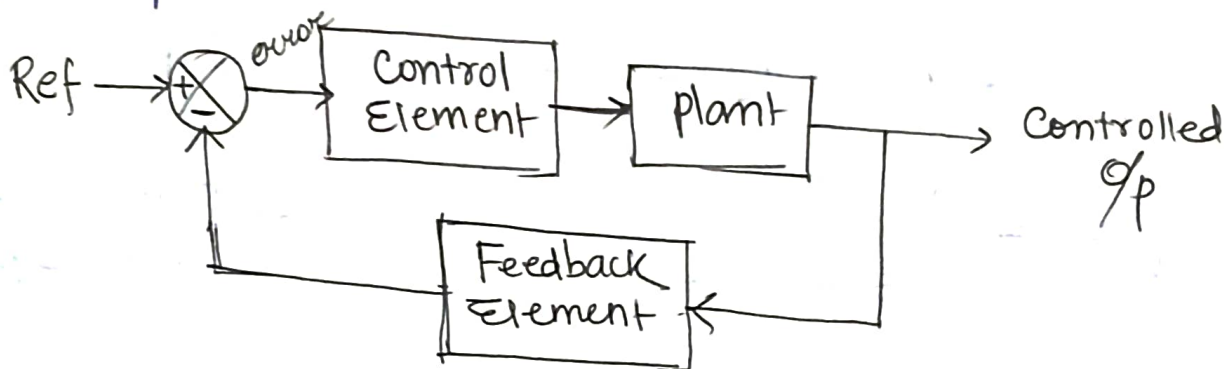
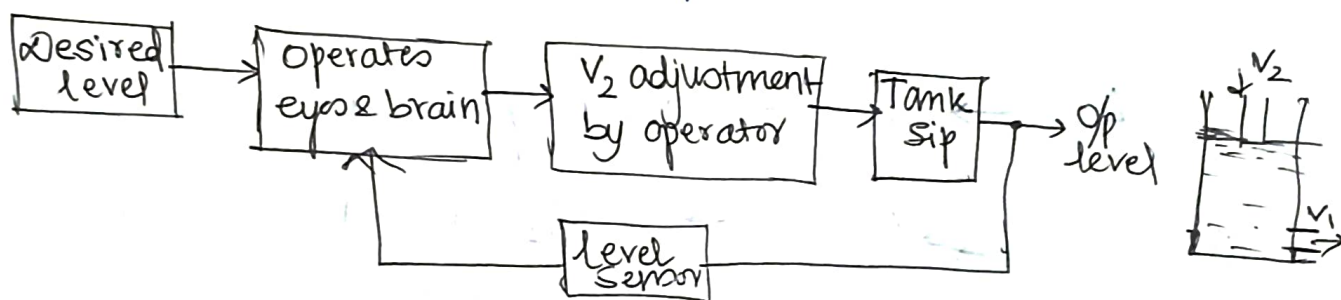


Control System:-

• Closed loop control:-



• Manually controlled closed-loop sys:



• System Models:-

1) Dynamic Eqn of any physical system is given by -

$$a_0 \frac{d^n y}{dt^n} + a_1 \frac{d^{n-1} y}{dt^{n-1}} + a_2 \frac{d^{n-2} y}{dt^{n-2}} + \dots + a_{n-1} \frac{dy}{dt} + a_n y = b_0 \frac{d^m r}{dt^m} + \dots + b_{m-1} \frac{dr}{dt} + b_m r$$

Where,

$y = y(t)$, o/p Variable

$r = r(t)$, i/p

$$n \geq m$$

System must be causal.

$a_i, b_j \rightarrow f^n$ of sys parameters.

→ The system o/p's are determined by past i/p's and not by future i/p's is called a Causal system.

e.g:- $n=0, m=1$

$$a_0 y = b_0 \frac{dr}{dt} + b_1 r, \text{ Let, } a_0 = b_0 = b_1 = 1 \rightarrow \text{future i/p not ca}$$

$$y = \frac{dr}{dt} + r \Rightarrow y(t) = r(t) + \frac{r(t+t_2) - r(t-t_2)}{2t}$$

2) Taking d.T with zero initial condⁿ:-

$$\Rightarrow s^n y(s) + a_1 s^{n-1} y(s) + \dots + a_n y(s) = b_0 s^m R(s) + \dots + b_m R(s)$$

$$\frac{Y(s)}{R(s)} = \frac{b_0 s^m + \dots + b_m}{s^n + \dots + a_n} = G(s) \leftarrow \text{Transfer function of the sys.}$$

$$G(s) = \frac{N(s)}{\Delta(s)} = \frac{K (s+z_1) (s+z_2) \dots (s+z_m)}{(s+p_1) (s+p_2) \dots (s+p_n)}$$

$\Delta(s) \rightarrow$ characteristic polynomial.

Characteristic Eqⁿ $\Rightarrow \Delta(s) = 0 \rightarrow s = -p_1, -p_2, -p_3, \dots, -p_n$ (poles)
 $N(s) = 0 \rightarrow s = -z_1, -z_2, \dots, -z_m$ (zeros)

* Transfer function of mass spring damper sys.

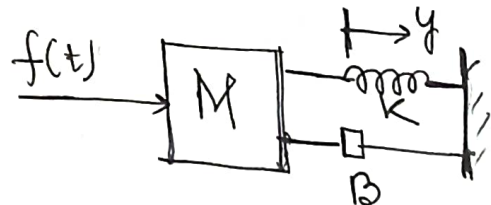
f : force

y : displacement

M : Mass

K : spring const.

B : damping co-efficient.



$$f(t) = Ky + B \frac{dy}{dt} + M \frac{d^2y}{dt^2}$$

Taking the d.T with zero initial condⁿ.

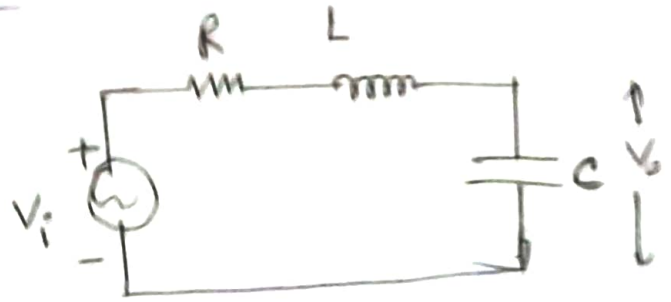
$$Ms^2 y(s) + B s y(s) + K y(s) = F(s)$$

$$\Rightarrow G(s) = \frac{Y(s)}{F(s)} = \frac{1}{Ms^2 + Bs + K}$$

* Transfer function of series RLC circ:

$$V_i = iR + L \frac{di}{dt} + V_o, \quad i = C \frac{dV_o}{dt}$$

$$\Rightarrow V_i = RC \frac{dV_o}{dt} + LC \frac{d^2 V_o}{dt^2} + V_o$$



d.T with zero initial condⁿ -

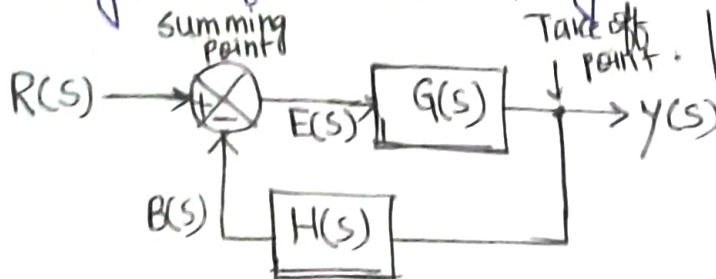
$$\Rightarrow V_i(s) = RC s V_o(s) + LC s^2 V_o(s) + V_o(s)$$

$$\Rightarrow \frac{V_o(s)}{V_i(s)} = \frac{1}{LC s^2 + RC s + 1} = \frac{1/LC}{s^2 + \frac{R}{L} s + \frac{1}{LC}}$$

* Block Diagram:-

$$Y(s) = G(s) R(s) \quad R(s) \longrightarrow \boxed{G(s)} \longrightarrow Y(s)$$

* Block diagram of a closed loop system:-



$R(s)$: Ref i/p

$Y(s)$: Controlled o/p

$G(s)$: Forward path TF

$H(s)$: Feedback path TF

$E(s)$: Error signal

$B(s)$: Feedback signal

$$Y(s) = G(s) E(s)$$

$$E(s) = R(s) - B(s)$$

$$= R(s) - H(s) Y(s)$$

$$\therefore Y(s) = G(s) [R(s) - H(s) Y(s)]$$

$$Y(s) = G(s) R(s) - G(s) H(s) Y(s)$$

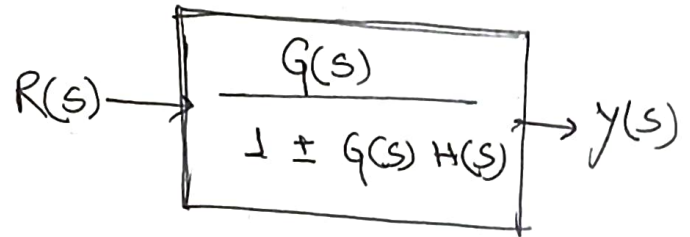
$$\Rightarrow [1 + G(s) H(s)] Y(s) = G(s) R(s)$$

$$Y(s) = \frac{G(s)}{1 + G(s)H(s)} R(s)$$

$$\Rightarrow \text{CL. T.f.} = \frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

General,

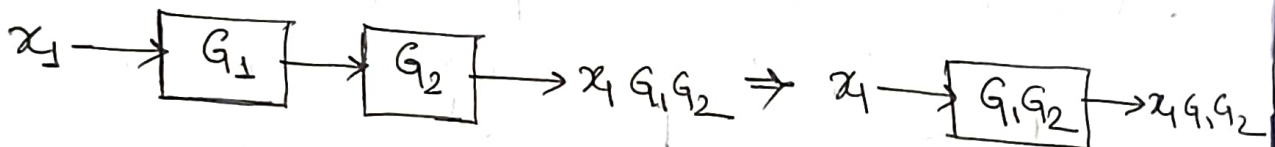
$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 \pm G(s)H(s)}$$



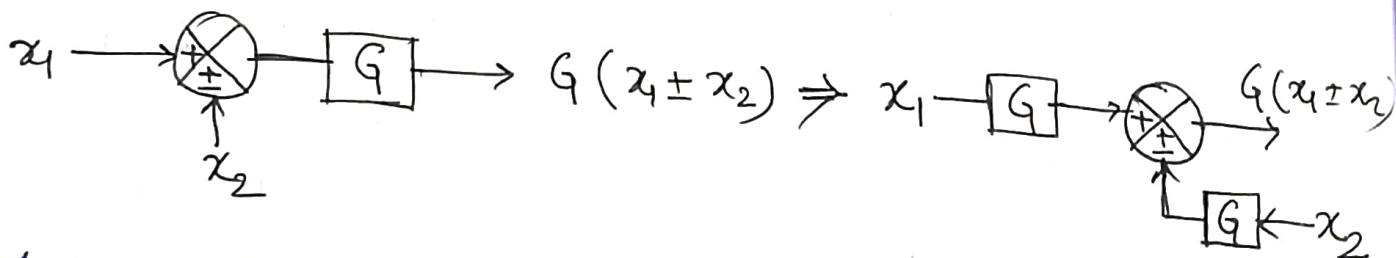
T.F for closed loop system.

* Block diagram reduction rule:

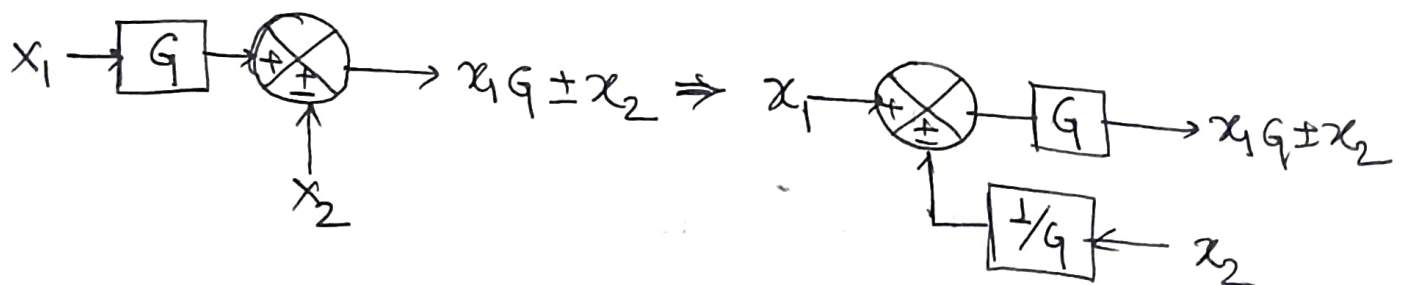
1. Combining blocks in cascade:



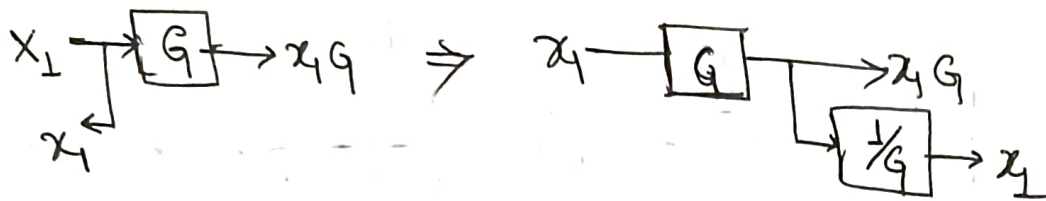
2. Moving a summing point after a block:



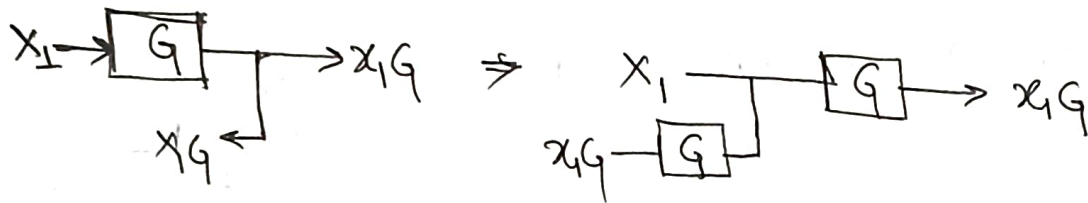
3. Moving a summing point ahead of a block:



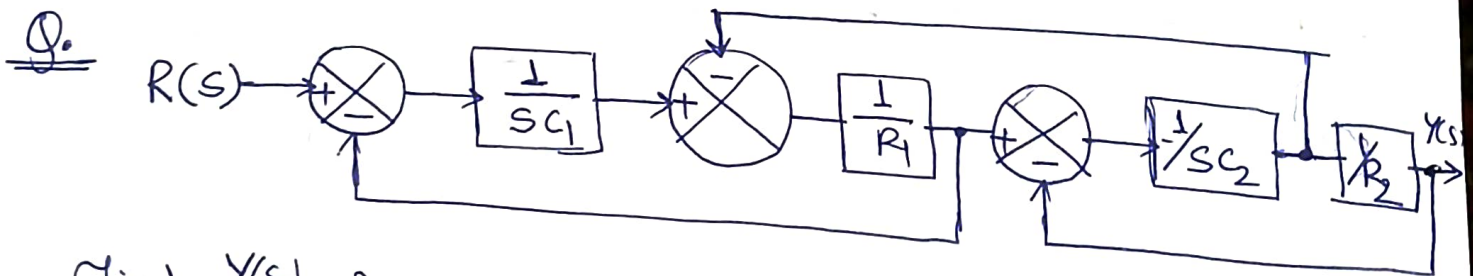
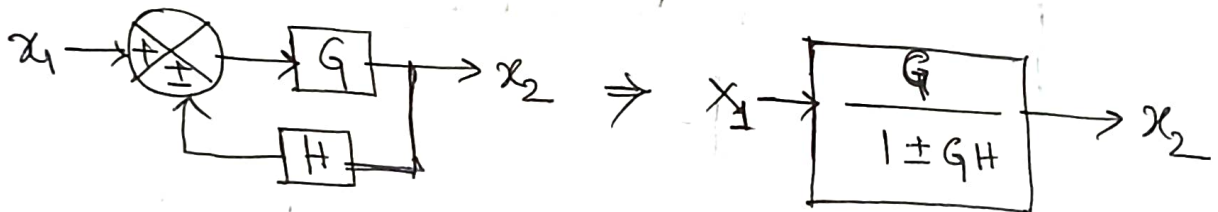
4. Moving a takeoff point after a block:-



5. Moving a take off point ahead of a block.



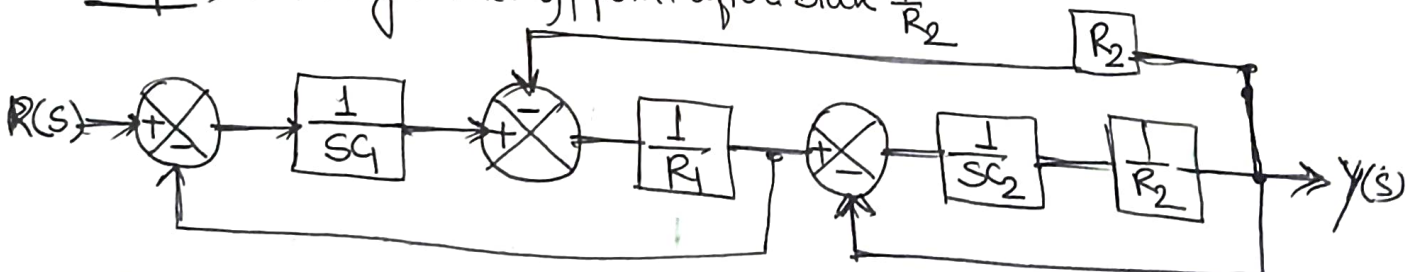
6. Eliminating feedback loop:



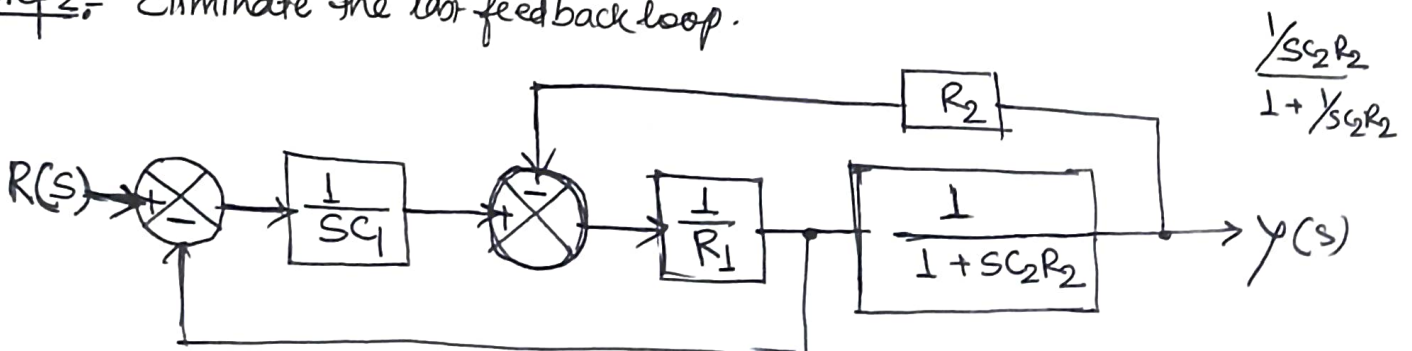
Find $\frac{Y(s)}{R(s)}$?

Soln:-

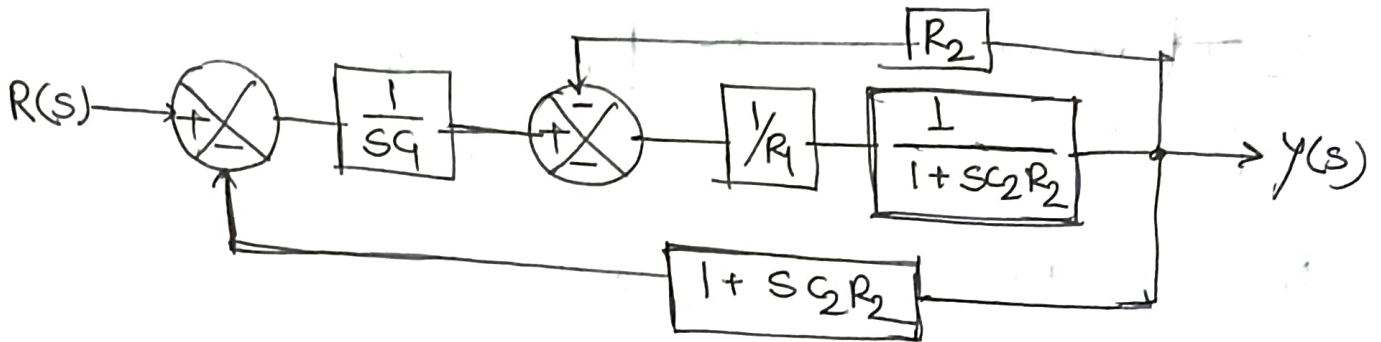
Step 1:- Moving a take-off point after block $\frac{1}{R_2}$



Step 2:- Eliminate the last feedback loop.

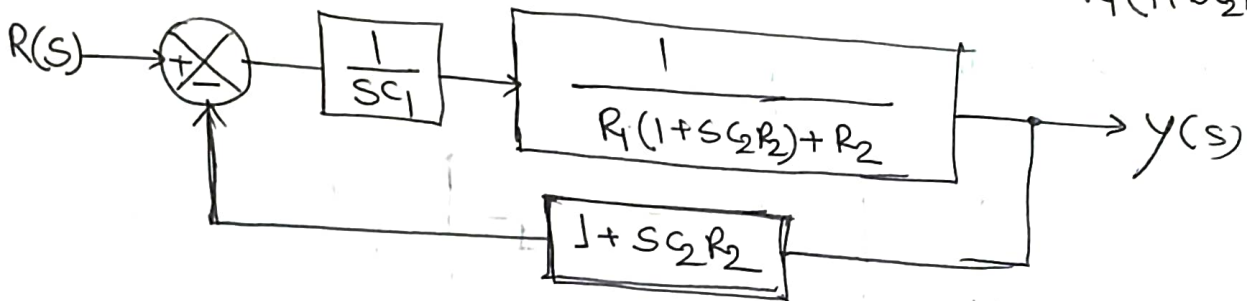


Step 3: Move the take-off point after the block $\frac{1}{1+sG_2R_2}$



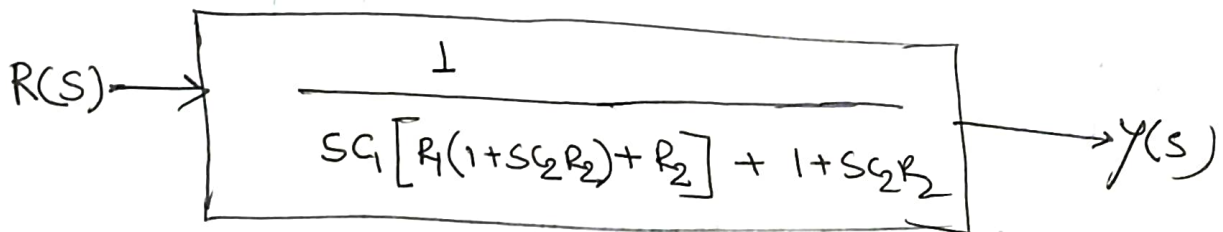
Step 4: Eliminate the feedback loop.

$$\frac{\frac{1}{R_1(1+sG_2R_2)}}{1 + \frac{R_2}{R_1(1+sG_2R_2)}}$$



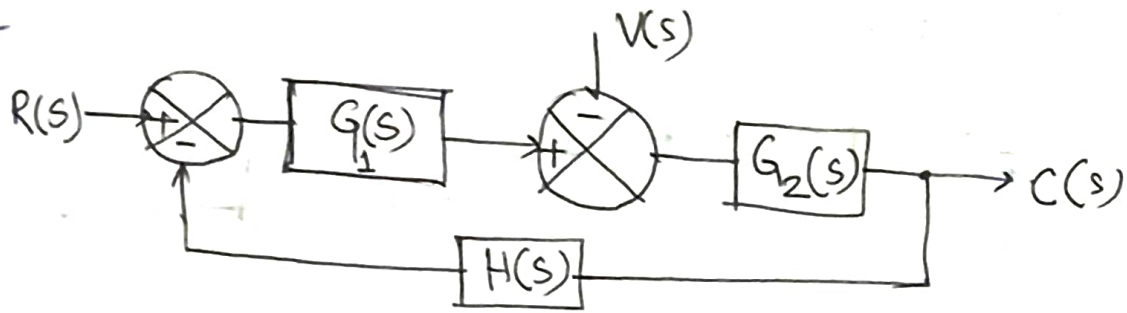
Step 5: Eliminate the feedback loop.

$$\frac{\frac{1}{sG_1} \frac{1}{R_1(1+sG_2R_2) + R_2}}{1 + \frac{1+sG_2R_2}{sG_1[R_1(1+sG_2R_2) + R_2]}}$$



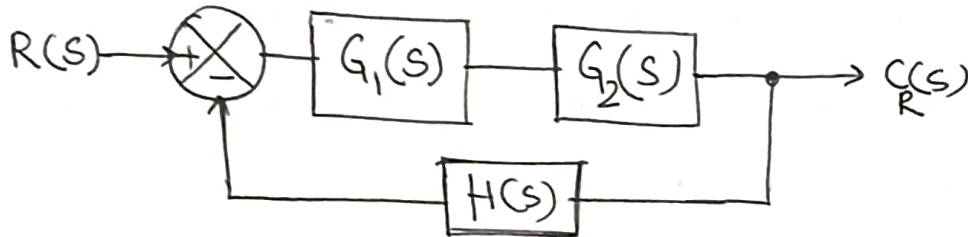
$$\therefore T(s) = \frac{Y(s)}{R(s)} = \frac{1}{sG_1 [R_1(1+sG_2R_2) + R_2] + 1+sG_2R_2}$$

eg:-



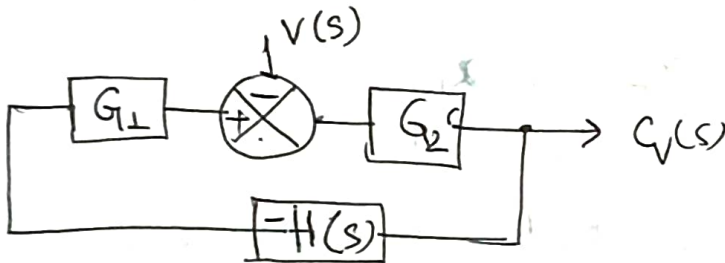
Soln:-

When, $V(s) = 0$



$$C_R(s) = \frac{G_1 G_2}{1 + G_1 G_2 H} R(s)$$

When $R(s) = 0$



$$C_V(s) = \frac{G_2 (-V(s))}{1 + G_1 G_2 H}$$

$$C_V(s) = \frac{-G_2 V(s)}{1 + G_1 G_2 H}$$

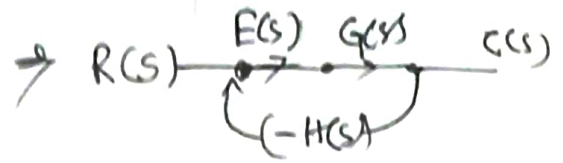
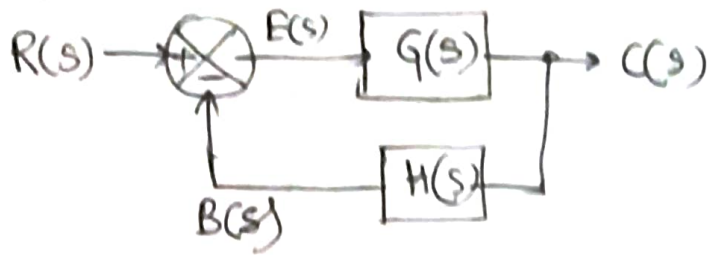
For linear systems superposition thm,

$$C(s) = C_R(s) + C_V(s)$$

$$= \frac{G_1 G_2 R(s)}{1 + G_1 G_2 H} + \frac{(-G_2 V(s))}{1 + G_1 G_2 H}$$

$$= \frac{G_2}{1 + G_1 G_2 H} [G_1 R(s) - V(s)]$$

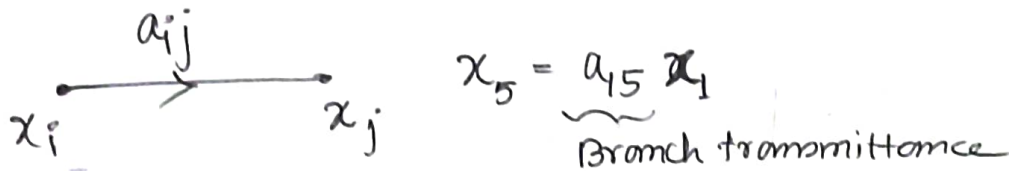
* Signal flow graph:-



1) Node:-

It represents a system variable which is required to the sum of all incoming signals at the node and outgoing signals do not effect the values of the node variable.

2) Branch:- Signal travels along a branch from one node to another in the direct indicated by the branch arrow and in the process gets multiplied by the gains or transmittance of the branch



a_{ij} is the transmittance of the branch directed from node x_i to x_j .

3) Input node or Source:- It is a node with only outgoing branches.

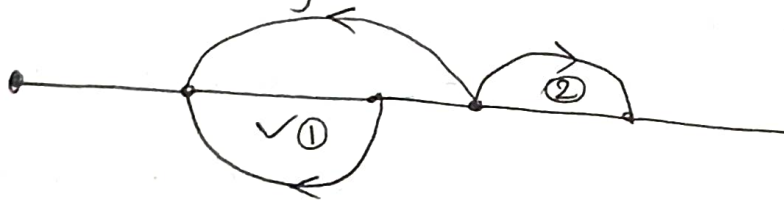
4) Output node:- It is a node with only incoming branches, however this condⁿ is not always met and additional branch with unity gain may be introduced in order to met this specified condⁿ.

5) Path:- It is the traversal of connected branch in the directⁿ of branch arrow, such that no node is traversed more than once.

6) Forward path:- It is a path from the i/p node to the o/p node.

7) Loop: It is a path which originates and terminates at the same node.

8) Non-touching loop:- Loops are said to be non touching if they do not share any common node.



① & ② non-touching

9) Forward path gain:- It is the product of the branch gain encountered in traversing a forward path.

10) Loop Gain:- It is a product of branch gain encountered in traversing the loop.

* Construction of Signal flow graph:

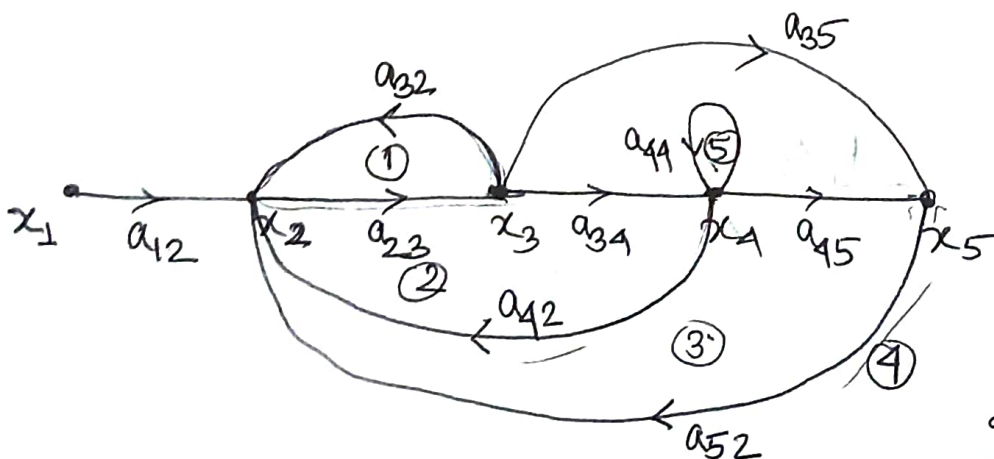
$$x_2 = a_{12}x_1 + a_{32}x_3 + a_{42}x_4 + a_{52}x_5$$

$$x_3 = a_{23}x_2$$

$$x_4 = a_{34}x_3 + a_{44}x_4$$

$$x_5 = a_{35}x_3 + a_{45}x_4$$

Where x_1 is the i/p variable and x_5 is the o/p variable



graph-1

* Mason's Gain Formula:-

$$T = \frac{1}{\Delta} \sum_k P_k \Delta_k$$

Where,

$k \rightarrow$ No. of forward path

$P_k \rightarrow$ path gain of k^{th} forward path

Δ_k = the value of Δ not touching with the k^{th} forward path

Δ = determinant of the graph

$$= 1 - (\text{sum of loop gains}) +$$

(sum of gain products of two non-touching loop)

+ (sum of ---)

Q. Find $\frac{x_5}{x_1}$? from the signal flow graph. 1

Soln, Here $k=2$

$$P_1 = a_{12} a_{23} a_{34} a_{45}$$

$$P_2 = a_{12} a_{23} a_{35}$$

$$\Delta = 1 - (a_{23} a_{32} + a_{23} a_{34} a_{42} + a_{23} a_{34} a_{45} a_{52} + a_{23} a_{35} a_{52} + a_{44}) + (a_{23} a_{32} a_{44} + a_{23} a_{35} a_{52} a_{44})$$

$$\Delta_1 = 1 - 0$$

$$\Delta_2 = 1 - a_{44}$$

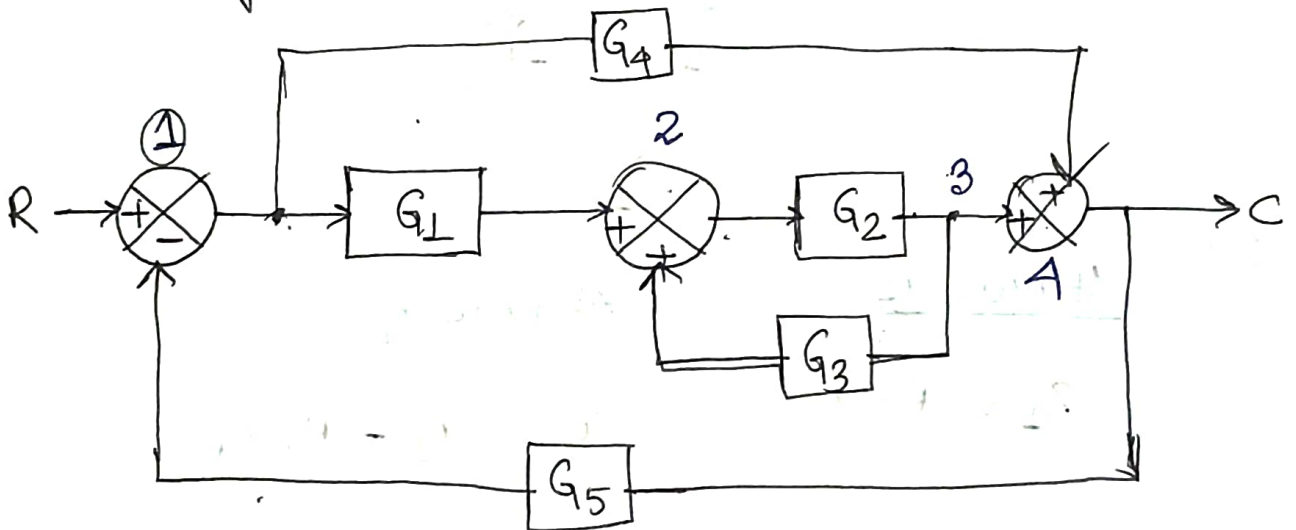
$$\therefore T = \frac{x_5}{x_1} = \frac{P_1 \Delta_1 + P_2 \Delta_2}{\Delta}$$

* Construction of Signal flow graph from Block diagram

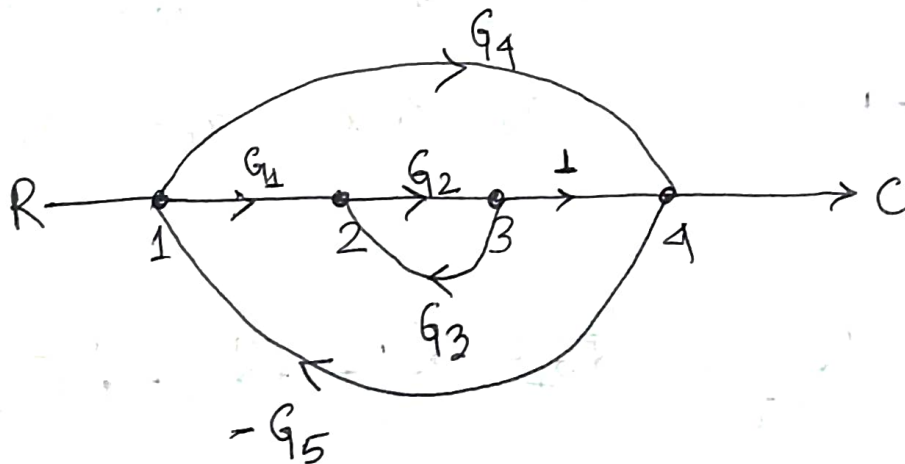
All variables ——— Summing points and takeoff points are represented by nodes

- ② If a summing point ^{is} ~~displaced~~ before a takeoff point in the direction of signal flow. In such case represent the summing point and take off point by a single ~~loop~~ node.
- ③ If a summing point is placed after a takeoff point in the direction of signal flow. In such case represent summing point and takeoff point by separate nodes connected by branch having transmittance ~~signal~~ unity.

e.g:-

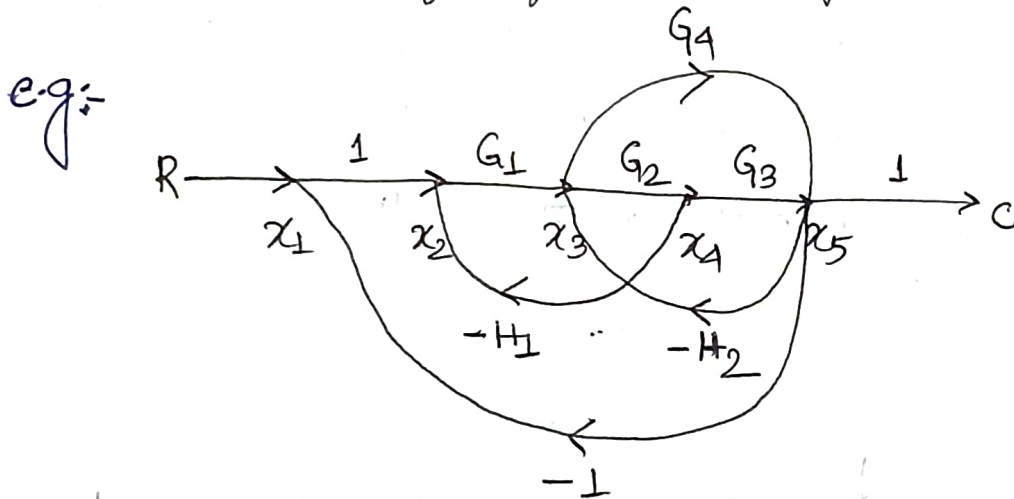


soln:-



* Block diagram from SFG:

- 1) For a given, SFG write system eqns.
- 2) At each node consider the incoming branches only.
- 3) Add all incoming signals algebraically at a node.
- 4) For positive & negative signs in system eqn, use a summing point.
- 5) For the gain of each branch of SFG draw the block having the same transfer function as the gain of the branch.



Soln:

At node x_1 ,

$$x_1 = R - x_5$$

At node x_2

$$x_2 = x_1 - H_1 x_4$$

At node x_3 ,

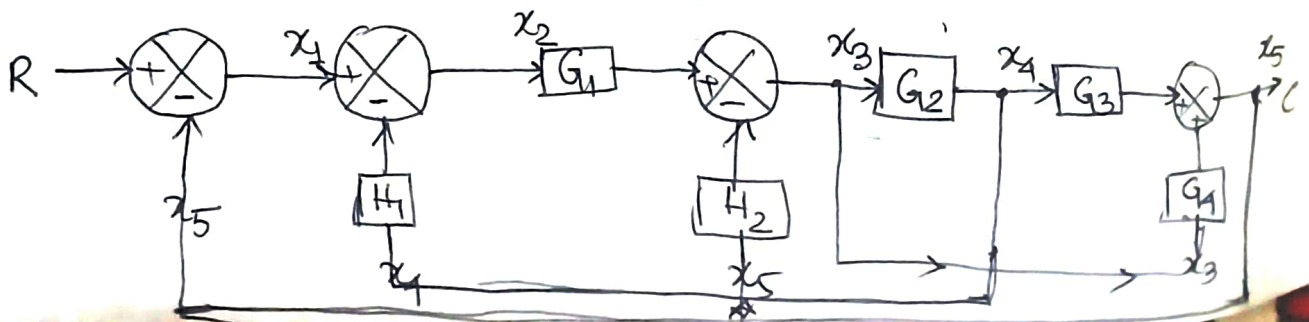
$$x_3 = G_1 x_2 - H_2 x_5$$

At node x_4

$$x_4 = G_2 x_3$$

At node x_5

$$x_5 = G_3 x_4 + G_4 x_3$$



* Time Response of first order System:

$$a_0 \frac{dy}{dt} + a_1 y = b_0 r$$

$y \Rightarrow \text{O/p}$

$r \Rightarrow \text{i/p}$

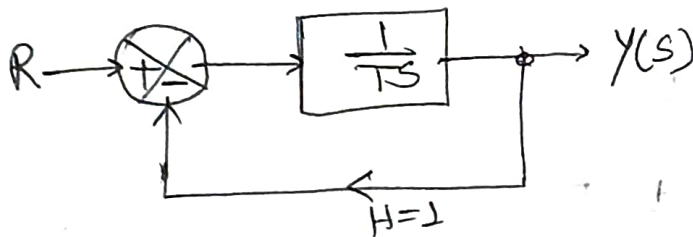
$$\Rightarrow \frac{a_0}{a_1} \frac{dy}{dt} + y = \frac{b_0}{a_1} r$$

$$\Rightarrow T \frac{dy}{dt} + y = K r$$

$$\Rightarrow sT Y(s) + Y(s) = K R(s)$$

$$\Rightarrow \frac{Y(s)}{R(s)} = \frac{K}{1 + sT}$$

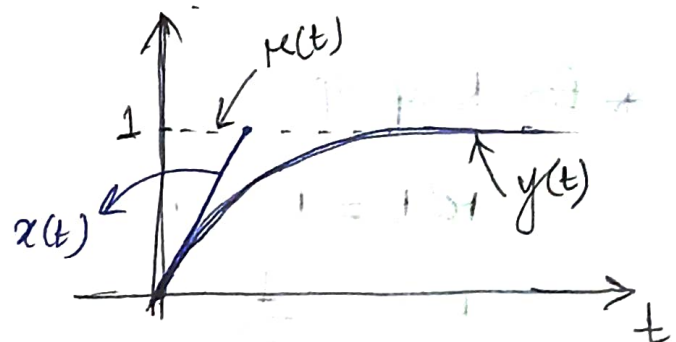
$$\left| \frac{\frac{K}{s}}{1 + \frac{K}{s}} \right| = \frac{1}{1 + sT}$$



$$\frac{Y(s)}{R(s)} = \frac{1}{1 + sT}$$

$\Rightarrow$ For unit step i/p.

$$r(t) = \begin{cases} 1 & ; t \geq 0 \\ 0 & ; t < 0 \end{cases}$$



$$R(s) = \frac{1}{s}, \quad s \neq 0$$

$$\therefore Y(s) = \frac{R(s)}{1 + sT} = \frac{1}{s(1 + sT)} = \frac{1}{s} - \frac{T}{1 + sT}$$

$$\therefore y(t) = (1 - e^{-t/T}) ; t \geq 0$$

$$\begin{aligned}\text{Error, } e(t) &= r(t) - y(t) \\ &= 1 - (1 - e^{-t/T}) \\ &= e^{-t/T}\end{aligned}$$

$$\text{Steady state Error} = \lim_{t \rightarrow \infty} e(t) = \lim_{t \rightarrow \infty} e^{-t/T} = 0$$

$$y(t) = (1 - e^{-t/T})$$

$$\left. \frac{dy}{dt} \right|_{t=0} = \left. \frac{e^{-t/T}}{T} \right|_{t=0} = \frac{1}{T}$$

Eqn of tangent :-

$$\cancel{y = \frac{1}{T} t} \quad x = \frac{1}{T} t$$

When $x=1$, $t=T$

Time constant is the time req^d to reach the steady state value if the initial rate is maintained.

* For Ramp i/p

$$r(t) = t, \quad t \geq 0$$

$$R(s) = \frac{1}{s^2}$$

$$Y(s) = \frac{R(s)}{1+Ts} = \frac{1}{s^2(1+Ts)} = \frac{A}{s^2} + \frac{B}{s} + \frac{C}{1+Ts}$$

$$y(t) = t - T(1 - e^{-t/T}); \quad t \geq 0$$

$$A = \lim_{s \rightarrow 0} \frac{d}{ds} \frac{1}{1+Ts}$$

$$= \lim_{s \rightarrow 0} \frac{-T}{(1+Ts)^2}$$

$$= -T$$

$$B = \left. \frac{1}{1+Ts} \right|_{s \rightarrow \infty} = 0$$

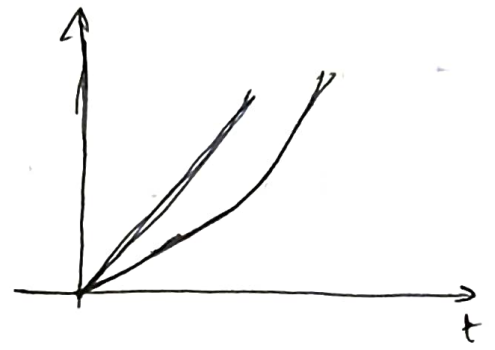
$$C = \left. \frac{1}{s} \right|_{s = -1/T} = -T$$

$$= \frac{1}{s^2} - T \left(\frac{1}{s^2} + \frac{1}{1+Ts} \right)$$

$$\text{Error, } e(t) = r(t) - y(t)$$

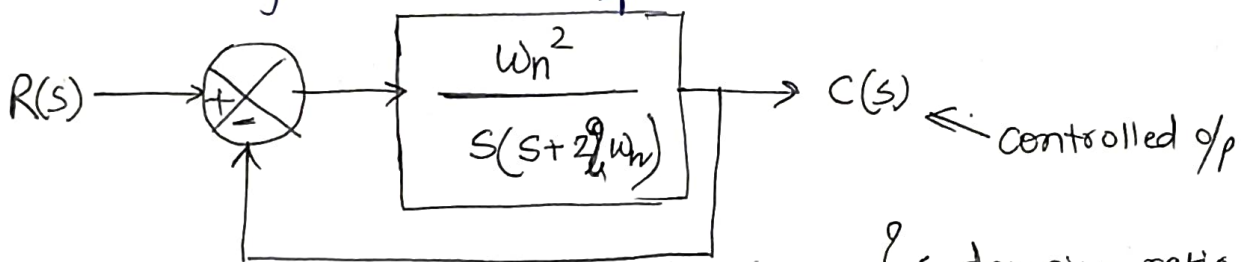
$$= t - (t - T(1 - e^{-t/T}))$$

$$= T(1 - e^{-t/T})$$



$$\therefore \text{Steady state error, } = \lim_{t \rightarrow \infty} T(1 - e^{-t/T}) = T$$

* Time Response of 2nd order system



ζ : damping ratio.

$$\text{Closed loop TF} = \frac{C(s)}{R(s)} = \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}$$

$$= \frac{\omega_n^2}{1 + \frac{\omega_n^2}{s(s + 2\zeta\omega_n)}}$$

ω_n : Undamped natural freqⁿ of oscillation.

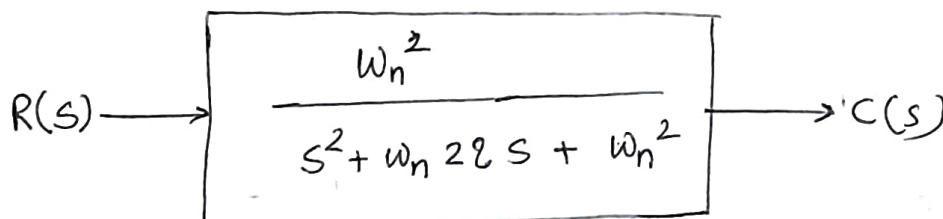
$$= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Forward path TF = G

Feedback TF = H

Loop TF = GH

CLTF = $\frac{G}{1 + GH}$



$$C(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} R(s)$$

* Unit Step Response:

$$r(t) = 1 ; t \geq 0$$

$$0 < t < 0$$

$$R(s) = \frac{1}{s}$$

$$\therefore C(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} R(s)$$

Nature of time response is decided by the roots of the characteristic Equation -

$$\text{Char. Eqn} \rightarrow s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\therefore s = \frac{-2\zeta\omega_n \pm \sqrt{(2\zeta\omega_n)^2 - 4\omega_n^2}}{2}$$

$$s_{1,2} = \cancel{-2\zeta\omega_n} - \zeta\omega_n \pm j\omega_n \sqrt{1 - \zeta^2}, \quad \boxed{\zeta < 1}$$

$$\therefore C(s) = \frac{\omega_n^2}{(s^2 + 2\zeta\omega_n s + \omega_n^2)} \cdot \frac{1}{s}$$

$$= \frac{\omega_n^2}{s(s-s_1)(s-s_2)}$$

$$C(s) = \frac{A}{s} + \frac{B}{s-s_1} + \frac{C}{s-s_2}$$

$$A = \left. \frac{\omega_n^2}{(s-s_1)(s-s_2)} \right|_{s=0} = \frac{\omega_n^2}{s_1 s_2}$$

$$B = \left. \frac{\omega_n^2}{s(s-s_2)} \right|_{s=s_1} = \frac{\omega_n^2}{s_1(s_1-s_2)}$$

$$C = \left. \frac{\omega_n^2}{s(s-s_1)} \right|_{s=s_2} = \frac{\omega_n^2}{s_2(s_2-s_1)}$$

Taking inverse Laplace T.

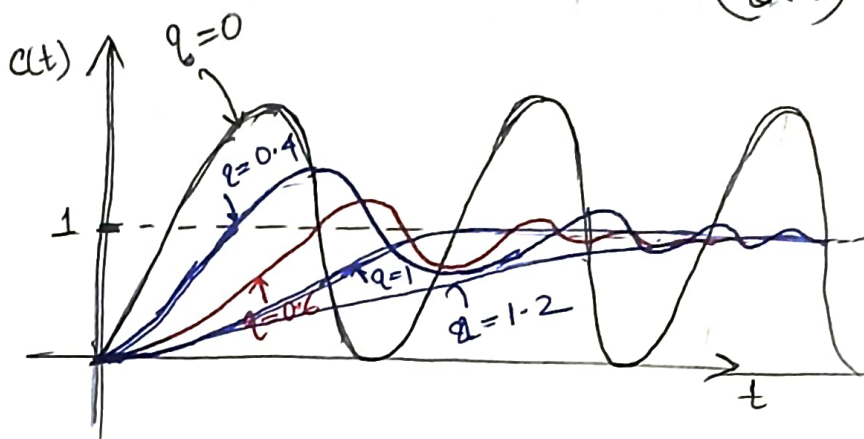
$$C(t) = A + B e^{s_1 t} + C e^{s_2 t}$$

After solving.

$$C(t) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin \left(\omega_n \sqrt{1-\zeta^2} t + \cos^{-1} \zeta \right); \quad t \geq 0 \quad (\zeta < 1)$$

When, $\zeta = 0$

$$C(t) = 1 - \cos \omega_n t$$

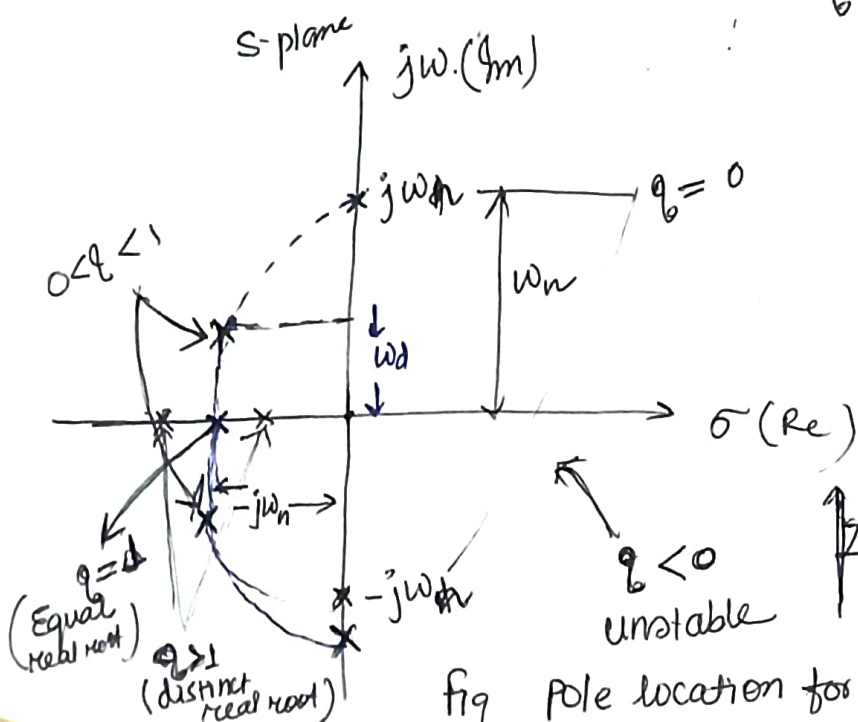


$$s_{1,2} = -\zeta \omega_n \pm j \omega_n \sqrt{1-\zeta^2}$$

$$= -\zeta \omega_n \pm j \omega_d$$

where $\omega_d = \omega_n \sqrt{1-\zeta^2}$
 $\downarrow$
 damped freqⁿ osc.

- $\zeta = 0 \rightarrow$ Undamped response. (purely oscillating)
- $0 < \zeta < 1 \rightarrow$ Underdamped " (damped sinusoid)
- $\zeta = 1 \rightarrow$ Critically damped " "
- $\zeta > 1 \rightarrow$ Overdamped "
- $\zeta < 0 \rightarrow$ ~~overdamped~~ unstable / negatively damped



When roots are on the $j\omega$ -axis $\rightarrow$ no damping
 $\rightarrow$ response is purely sinusoidal

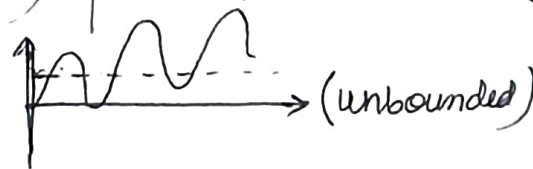


Fig Pole location for a second order eqn for different ζ

eg.



$$(1 - e^{-t})$$



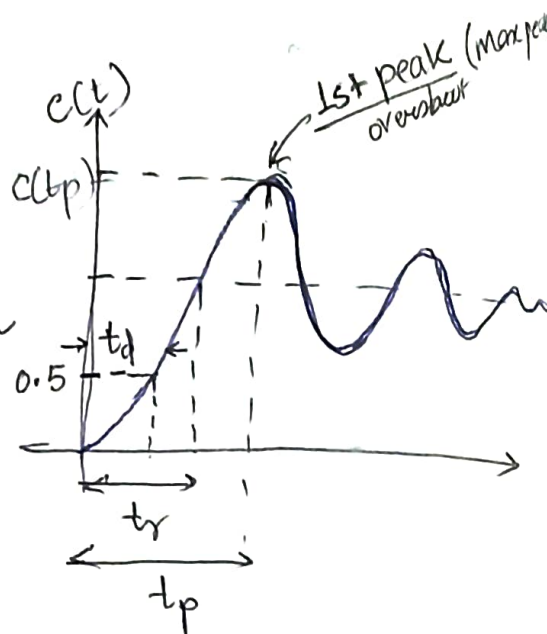
2nd order
critically or overdamped

* Time response Specification :-

1. Delay time (t_d) Time ^{required} response to reach 50% of steady state value.

2. Rise time (t_r) :- Time required to reach 10-90% of steady state value for overdamped sys.

0-100% of the steady state value for underdamped sys.



$$C(t_r) = 1 - \frac{e^{-\zeta \omega_n t_r}}{\sqrt{1 - \zeta^2}}$$

$$\sin(\omega_n \sqrt{1 - \zeta^2} t_r + \cos^{-1} \zeta) = 1$$

$$t_r = \frac{\pi - \cos^{-1} \zeta}{\omega_n \sqrt{1 - \zeta^2}}$$

→ Rise time for $0 < \zeta < 1$

3. Peak time (t_p) Time reqd to reach maximum of peak value is called peak time.

$$C(t) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_n \sqrt{1-\zeta^2} t + \cos^{-1} \zeta)$$

$$\frac{dC(t)}{dt} = 0$$

After Solving, General solution,

$$t_p = \frac{n\pi}{\omega_n \sqrt{1-\zeta^2}} ; n=0,1,2,\dots$$

When, $n=1$ (1st overshoot) , when $n=2$ $\rightarrow$ (1st undershoot)

$$t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}$$

$$t_2 = \frac{2\pi}{\omega_n \sqrt{1-\zeta^2}}$$

4. Max peak overshoot (M_p):-

$$M_p = C(t_p) - 1$$

$$= \cancel{e^{-\pi\zeta/\sqrt{1-\zeta^2}}} e^{-\pi\zeta/\sqrt{1-\zeta^2}}$$

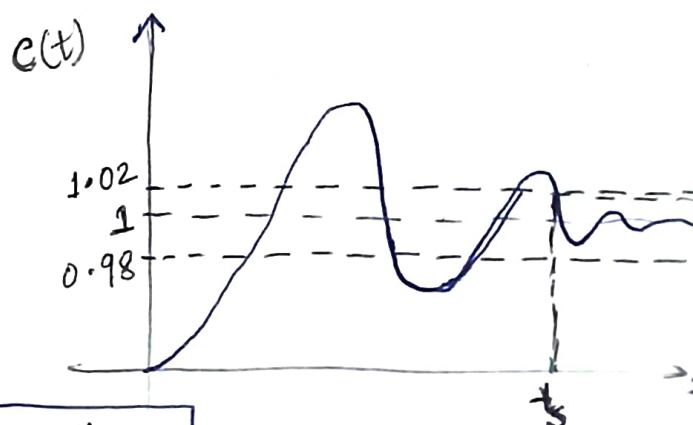
$$\% \text{ Maximum peak overshoot} = e^{-\pi\zeta/\sqrt{1-\zeta^2}} \times 100\%$$

5. Settling time (t_s):-

$$1 \pm \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} = 1 \pm 2\%$$

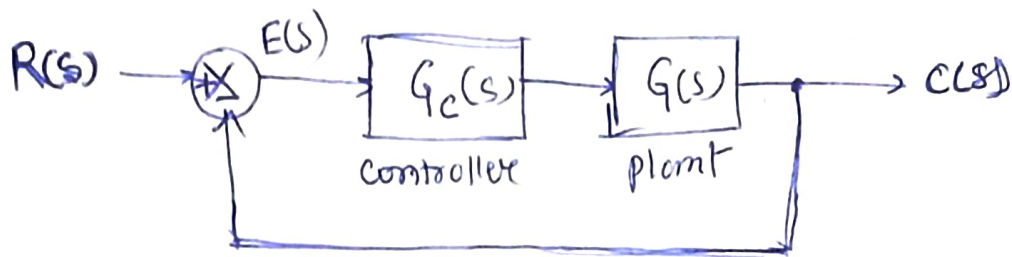
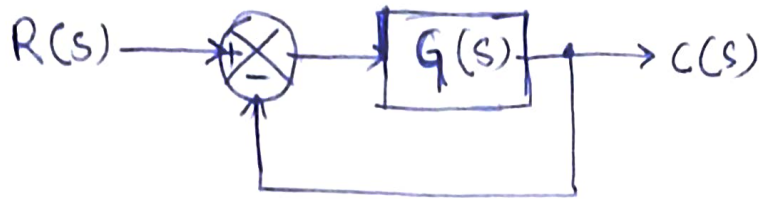
$$\frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} = 0.02$$

$$\Rightarrow e^{-\zeta \omega_n t} = 0.02 \Rightarrow t_s = \frac{4}{\zeta \omega_n}$$

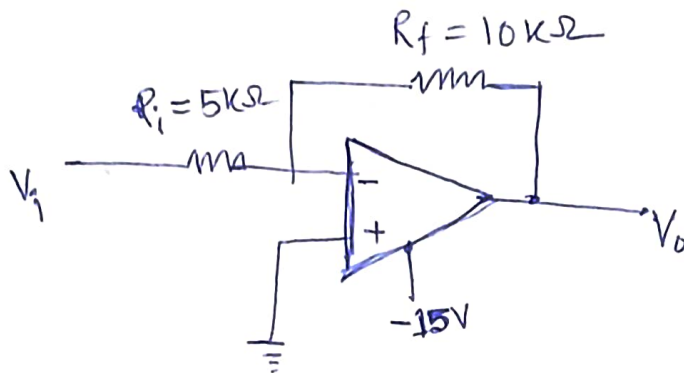


* Steady state error:-

$$E(s) = R(s) - C(s)$$

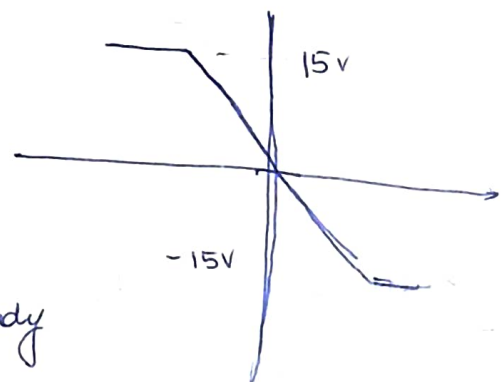


Assume that G_c uses an amplifier,



$$\begin{aligned} V_o &= -\frac{R_f}{R_i} V_i \\ &= -\frac{10}{5} V_i \\ &= -2V_i \end{aligned}$$

V_i	V_o
1	-2
2	-4
3	-6
4	-8
5	-10
10	-15



Non linearity of the controller causes steady state error

controlled q/p, $C(s) = \frac{G(s)}{1+G(s)} R(s)$

$$C(s) = E(s) G(s)$$

$$\therefore G(s) E(s) = \frac{G(s)}{1+G(s)} R(s)$$

$$E(s) = \frac{R(s)}{1+G(s)}$$

* Steady state error:-

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} s E(s), \text{ where } E(s) = \mathcal{L}\{e(t)\}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s(R(s))}{1+G(s)}$$

, for linear system, steady state error depends on loop gain.

* Unit step input:-

$$r(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{s}$$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} s E(s)$$

$$= \lim_{s \rightarrow 0} \frac{s R(s)}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{1+G(s)} = \frac{1}{1+G(0)}$$

$$= \frac{1}{1+K_p}$$

where $K_p = \lim_{s \rightarrow 0} G(s) \rightarrow$ position error const.

* Ramp i/p

$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{s^2}$$

$$e_{ss} = \lim_{s \rightarrow 0} s R(s) \frac{1}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s + s G(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} s G(s)} = \frac{1}{K_V}$$

where, $K_V = \lim_{s \rightarrow 0} s G(s)$

Velocity error const.

* parabolic i/p

$$r(t) = \begin{cases} \frac{t^2}{2}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{2s^3}$$

$$e_{ss} = \lim_{s \rightarrow 0} s R(s) \frac{1}{1+G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} s^2 G(s)}$$

$$= \frac{1}{K_A}$$

where

$$K_A = \lim_{s \rightarrow 0} s^2 G(s)$$

= Acceleration error const.

* Consider the - dTF

$$G(s) = \frac{\alpha (T_{z1}s + 1) (T_{z2}s + 1) \dots}{s^p (T_{p1}s + 1) (T_{p2}s + 1) \dots} \quad \left(\begin{array}{l} \text{Time-constant} \\ \text{form} \end{array} \right)$$

type no. $\rightarrow$ No. of poles
at origin
 $= p$

$$= \frac{K' (s + T_1) (s + T_2) \dots}{s^p (s + P_1) (s + P_2) \dots} \quad \left(\begin{array}{l} \text{pole-zero form} \end{array} \right)$$

where $K' = K \frac{\pi_i T_{zi}}{\pi_i T_{pi}}$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)}, \text{ Steady state error is dominated by the no. of poles at origin.}$$

Type-0 sys:- , $n=0$

Unit step

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$U(s) = \frac{1}{s}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s R(s)}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{1 + G(s)}$$

$$= \frac{1}{1 + K}$$

Ramp i/p

$$u(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$R(s) = \frac{1}{s^2}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \frac{1}{s^2}}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s + s G(s)}$$

$$= \frac{1}{\lim_{s \rightarrow 0} s G(s)} = \frac{1}{K_v} = \infty$$

* Parabolic i/p

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \times \frac{1}{s^3}}{1 + G(s)}$$

$$= \lim_{s \rightarrow 0} \frac{1}{s^2 + s^2 G(s)}$$

$$= \infty$$

Type-n

$$G(s) = \frac{K (T_{z1}s+1) (T_{z2}s+1) \dots}{s^n (T_{p1}s+1) (T_{p2}s+1) \dots}$$

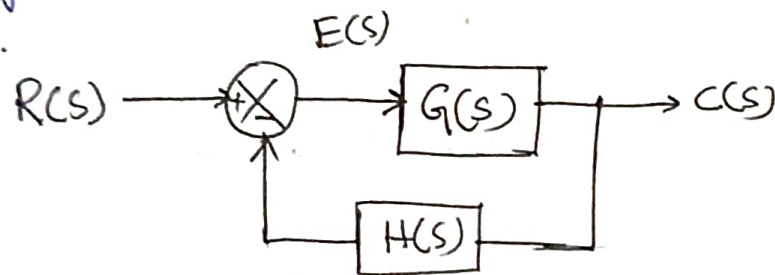
	Type-0	Type-1	Type-2
Step i/p	$\frac{1}{1+K}$	0	0
Ramp i/p	∞	$\frac{1}{K}$	0
Parabolic i/p	∞	∞	$\frac{1}{K}$

→ As the ~~steady~~ type no. (n) increases, steady state error reduces.

→ If we go beyond 2 (typeno.), steady state error will reduce by dynamic error increases and difficult to stabilize.

→ Also we will not get information, how the steady state error varies with time.

* Steady state error for non-unity feedback control system.



$$C(s) = \frac{G(s)}{1 + G(s)H(s)} R(s) \quad \rightarrow (1)$$

$$E(s) = R(s) - C(s)H(s)$$

$$\therefore C(s) = E(s)G(s)$$

$$(1) \Rightarrow E(s) = \frac{R(s)}{1 + G(s)H(s)}$$

Steady state error,

$$E_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \frac{R(s)}{1 + G(s)H(s)}$$

Q. For a unit feedback control system, the forward path transfer function is given by —

$$G(s) = \frac{20}{s(s+2)(s^2+2s+20)}, \text{ Determine the}$$

steady state error of the system, when the i/p's are

(i) 5 (ii) 5t (iii) $\frac{3t^2}{2}$

Soln:

$$e_{ss} = \lim_{s \rightarrow 0} \frac{sR(s)}{1+G(s)}$$

(i) $R(s) = \frac{5}{s}$

$$\therefore e_{ss} = \lim_{s \rightarrow 0} \frac{s \times \frac{5}{s}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= 0$$

(ii) $R(s) = \frac{5}{s^2}$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{\frac{5}{s}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= \lim_{s \rightarrow 0} \frac{5}{s + \frac{20}{(s+2)(s^2+2s+20)}}$$

$$= \frac{5}{\frac{20}{2 \times 20}} = \frac{5}{2}$$

(iii) $R(s) = \frac{3}{2s^3}$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s \frac{3}{2s^3}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= \lim_{s \rightarrow 0} \frac{\frac{3}{2s^2}}{1 + \frac{20}{s(s+2)(s^2+2s+20)}}$$

$$= \lim_{s \rightarrow 0} \frac{\frac{3}{2}}{\frac{s^2 + \frac{20s}{(s+2)(s^2+2s+20)}}} = \infty$$

Q A 2nd order sys. has transfer function given by - (GATE 1991)

$$G(s) = \frac{25}{s^2 + 8s + 25} \text{ . if the sys. initially at rest}$$

is subjected to a unit step input at $t=0$, the 2nd peak in the response will occur at ~~5 sec~~ (a) π sec (b) $\pi/2$ sec (c) $2\pi/3$ sec (d) $\pi/3$ sec

Soln: Stand. 2nd order sys. T.F. -

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\therefore \omega_n = 5 \text{ Rad/s}$$

$$2\zeta\omega_n = 8$$

$$\Rightarrow \zeta = \frac{8}{2 \times 5} = 0.8$$

$$\text{Now, } t_p = \frac{n\pi}{\omega_d} = \frac{n\pi}{\omega_n \sqrt{1-\zeta^2}} \quad \checkmark$$

For 2nd peak, $n=3$

$$t_p = \frac{3\pi}{5 \sqrt{1-(0.8)^2}} \text{ sec} = \pi \text{ sec}$$

Q The open loop TF of a unity feedback sys. is given by -

$$G(s) = \frac{K}{s(1+st)} \text{ , where, } K \text{ and } T \text{ are positive const. By}$$

what factor should the amplifier gain be reduced so that the peak overshoot of unit step response of the sys. is reduced from 75% to ~~75~~ 25%.

Soln.:

$$\text{Closed loop TF. } \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$= \frac{\frac{K}{s(1+sT)}}{1 + \frac{K}{s(1+sT)}} = \frac{K}{s^2T + s + K} = \frac{K/T}{s^2 + \frac{1}{T}s + K/T}$$

$$\text{Here } \omega_n^2 = \frac{K}{T}$$

$$\Rightarrow \omega_n = \sqrt{\frac{K}{T}} \text{ rad/sec}$$

$$2\zeta\omega_n = \frac{1}{T}$$

$$\Rightarrow \zeta = \frac{1}{2T \times \sqrt{\frac{K}{T}}} = \frac{1}{2\sqrt{KT}}$$

$$M_p = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

$$\text{Now, } e^{-\frac{\pi\zeta_1}{\sqrt{1-\zeta_1^2}}} = 0.75$$

$$\Rightarrow \zeta_1 = 0.09118$$

$$e^{-\frac{\pi\zeta_2}{\sqrt{1-\zeta_2^2}}} = 0.25$$

$$\Rightarrow \zeta_2 = 0.4037$$

$$\therefore \frac{\zeta_1}{\zeta_2} = \frac{\frac{1}{2\sqrt{KT}}}{\frac{1}{2\sqrt{K_2T}}} = \sqrt{\frac{K_2}{K_1}} = \frac{0.09118}{0.4037}$$

$$\boxed{K_1 = 19.6 K_2}$$

Stability:-

An LTI system is stable if the following two notions of stability are satisfied -

- ① If i/p is bounded, output is bounded (BIBO)
- ② If there is no i/p, o/p tends towards zero
- asymmetric stability.

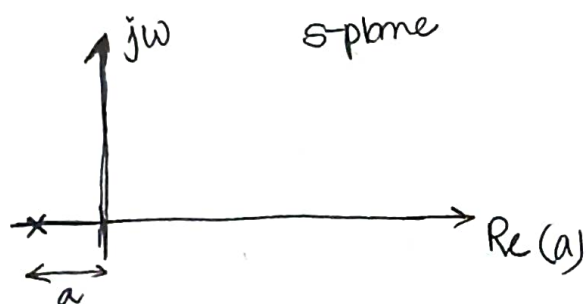
$$\frac{d^2 y}{dt^2} + 2 \zeta \omega_n \frac{dy}{dt} + \omega_n^2 y = x(t)$$

Unbounded i/p $\Rightarrow$ unbounded o/p.

Then nothing can be said about such sys.

depends on -
initial condⁿ
sys. parameters
Small change in the
i/p large change in o/p

Pole position:-

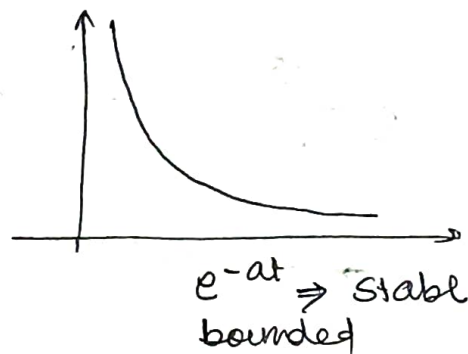


$$\frac{C(s)}{R(s)} = \frac{1}{s+a}$$

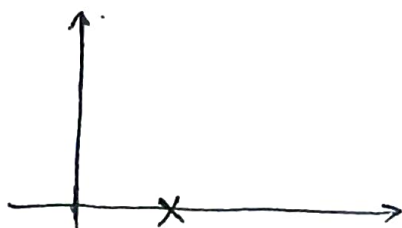
$$C(s) = \frac{1}{s+a}$$

$$\Rightarrow c(t) = e^{-at}$$

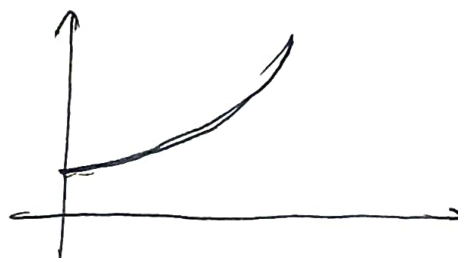
Impulse response

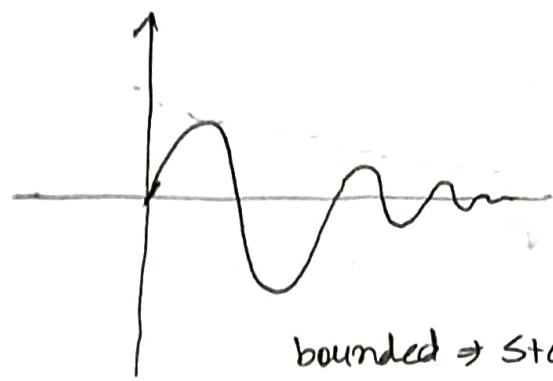
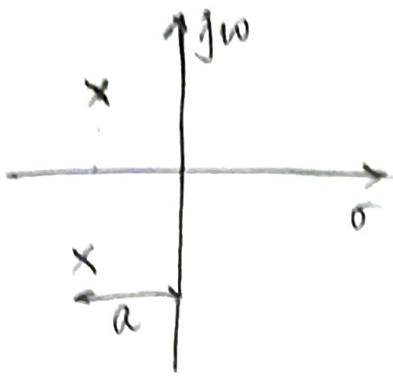


Pole position:-



Impulse response



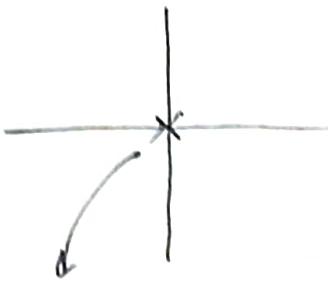


bounded $\Rightarrow$ stable

TF: $\frac{1}{(s+a)^2 + w_n^2}$

$e^{-at} \sin w t$

$c(t) = e^{-at} \sin w t$

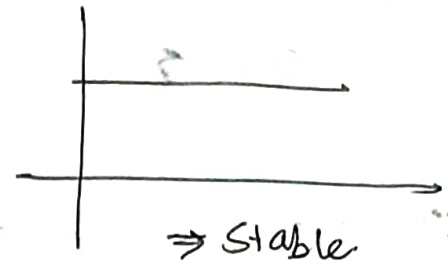


TF = $\frac{1}{s}$

$C(s) = \frac{1}{s} R(s)$

$= \frac{1}{s} \cdot \frac{1}{s+1}$

$= \frac{1}{s} - \frac{1}{s+1}$



$\Rightarrow$ Stable

marginally stable;

root posⁿ same

for different $c(t) = (1 - e^{-t}) u(t)$

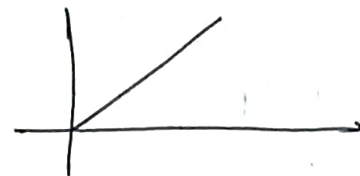
subjected signal

diff o/p; et or unit.

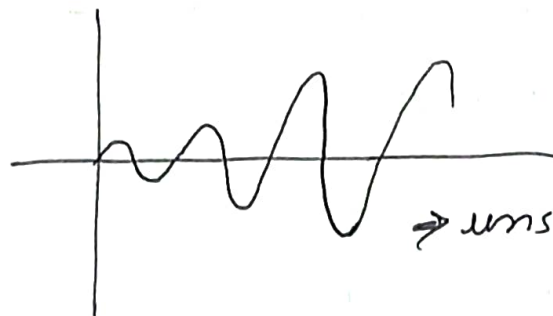
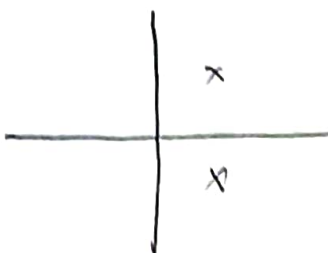
Now, subjected to unit step.

$C(s) = \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2}$

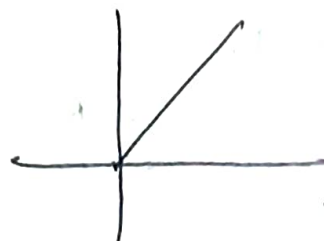
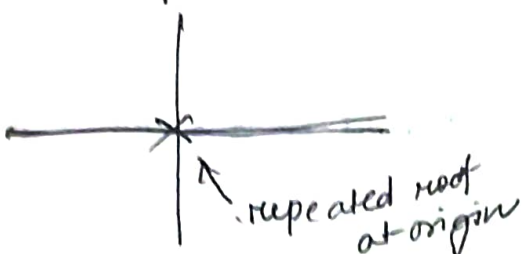
$c(t) = t u(t)$



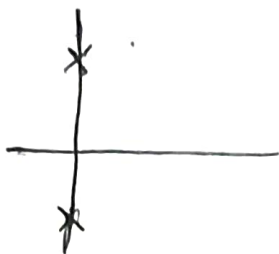
unstable.



$\Rightarrow$ unstable

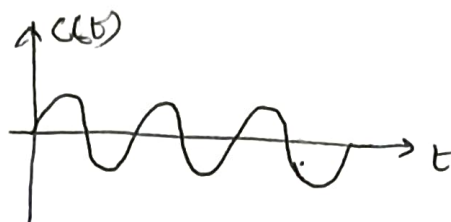


$\Rightarrow$ unstable

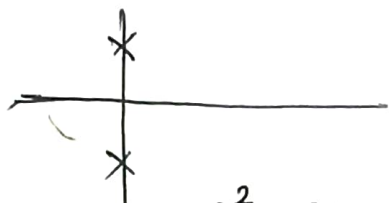


$$C(s) = \frac{1}{s^2 + \omega^2}$$

$$c(t) = \sin \omega t$$

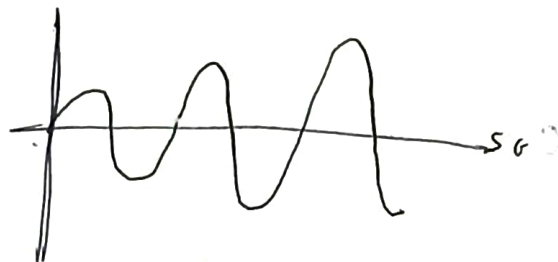


⇒ Stable
(marginally stable)



$$s^2 + 2\zeta \omega_n s + \omega_n^2 = 0$$

Characteristic eqn.



⇒ Unstable

Roots → left side → stable

Roots → on $j\omega$ → single pair → marginally stable.

Roots → on right → unstable.

Roots → on $j\omega$ (repeated) → unstable.

→ Conclusion: poles with negative real part stable system.

poles with positive real part unstable.

One or more non-repeated poles on $j\omega$ → marginally stable or limited by stable.

Repeated poles on $j\omega$ → unstable.

* Necessary conditions for stability:-

$$\Phi(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0 = 0$$

all the co-eff. should be of same sign (real) no. co eff. should be zero.

$$s^2 - 2s + 1 = 0 \Rightarrow \text{co-eff. different sign} \Rightarrow \text{unstable.}$$

* Sufficient condⁿ:-

$$G(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0 = 0$$

first column elements of Routh array should be of same sign for the sys. will be stable.

if any sign changes → unstable

no. of sign changes indicated no. of roots in the right half of s-plane.

* Routh stability criterion:

$$G(s) = a_0 s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_n = 0$$

s^n	a_0	a_2	a_4
s^{n-1}	a_1	a_3	a_5
s^{n-2}	b_1	b_3	b_5
$\vdots$	$\vdots$	$\vdots$	$\vdots$
s^0	$\vdots$	$\vdots$	$\vdots$

$$b_1 = \frac{a_1 a_2 - a_0 a_3}{a_1}$$

$$b_2 = \frac{a_1 a_4 - a_0 a_5}{a_1}$$

first column
if same sign
→ stable

$+a_0$
 $+a_1$
 $-b_1$
 $+b_2$
 $\vdots$

sign change = 2, means 2 roots in right side of s-plane.

Prob 1:- $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$

The Routh Table:

s^4	1	18	5
s^3	8	16	
s^2	$\frac{18-2}{1} = 16$	$\frac{5-0}{1} = 5$	
s^1	$2\frac{7}{16}$	0	
s^0	5		

no sign change → stable

any row can be divided by common factor

Prob 2:- $3s^4 + 10s^3 + 5s^2 + 5s + 2 = 0$

s^4	3	5	2
s^3	10	5	0
s^2	$\frac{50-15}{10} = \frac{7}{2}$	2	
s^1	$-\frac{1}{7}$	0	
s^0	2	0	

There are two sign changes

+ve → -ve

-ve → +ve

∴ unstable system (with two poles in Right hand s-plane)

Here is a sign change → unstable
There are two sign changes → two roots in Right side s-plane

Special Cases:

- (i) When the 1st term in any row is zero, while rest of the row have atleast one non-zero term. (It has to first fulfill the necessary condition)

$$q(s) = s^n + () s^{n-p} + () \dots$$

$$\begin{array}{c|ccc} s^n & 1 & 2 & \dots \\ s^{n-p} & 1 & 2 & \dots \\ \vdots & 0 & \dots & \dots \\ \hline & -2 \left(\frac{2\epsilon+2}{\epsilon} \right) - 5\epsilon & & \\ & \Rightarrow -4 - \frac{4}{\epsilon} - 5\epsilon & & \\ & \frac{2\epsilon+2}{\epsilon} & & \end{array}$$

eg:- Consider the characteristic equation,

$$s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$$

The Routh array is -

$$\begin{array}{c|ccc} s^5 & 1 & 2 & 3 \\ s^4 & 1 & 2 & 5 \\ s^3 & \frac{2\epsilon+2}{\epsilon} & -2 & \\ s^2 & -4 - \frac{4}{\epsilon} - 5\epsilon & \frac{2\epsilon+2}{\epsilon} & 5 \\ s^1 & -\frac{4\epsilon-4-5\epsilon^2}{2\epsilon+2} & (-) & \\ s^0 & 5 & (+) & \end{array}$$

NOTE: If any one of the element is zero, then, s^2 elements will be undefined.

Because of this zero term, in the next row, the terms becomes infinite and Routh test breakdown.

Method: Substitute a small positive no. (ϵ) for the zero, and proceed to evaluate the rest of the Routh array, then examine the sign of the first column of Routh array by making ϵ tends to zero.

there is two sign changes ($+\rightarrow-\rightarrow+$)

$\therefore$ The system is unstable, with two poles in the right-half s-plane.

(ii) All elements of a particular row is zero.

* Row

Note:- When all the elements in any one row of the Routh array are zero, this indicates symmetrically located roots in the s-plane (pair of real roots with opposite signs and/or pair of conjugate roots on the imaginary axis and/or complex conjugate roots forming quadratically in the s-plane).

The polynomial whose co-efficients are the elements of the row just above the row of zeros in the Routh array is called an auxiliary polynomial.

The auxiliary polynomial gives the no. and locations of root pairs, which are symmetrically located. The order of auxiliary polynomial is always even.

* Relative stability Analysis -

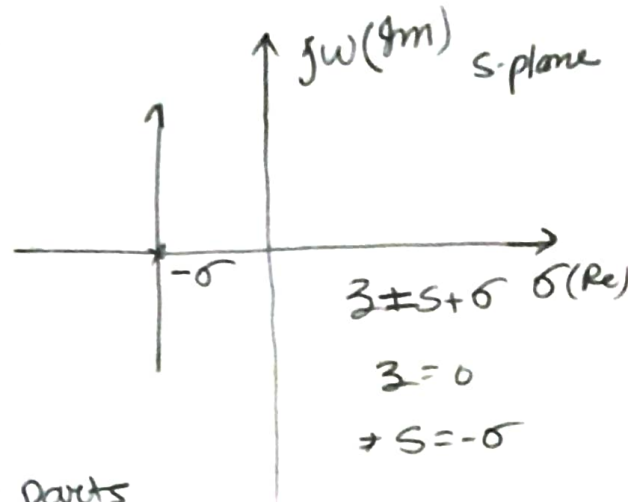
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The relative stability can be specified by ~~the~~ enquiring

all the roots of the characteristic eqn which is more negative of the certain values. i.e all the roots must lie on the left of the s-plane ($s = -s_1, s_1 > 0$)

Q. Consider the following characteristic

Eqn: $s^3 + 7s^2 + 25s + 39 = 0$



Let us check if all the roots have real parts more negative than -1.

Soln:

$$z = s + 1.$$

$$\Rightarrow s = z - 1$$

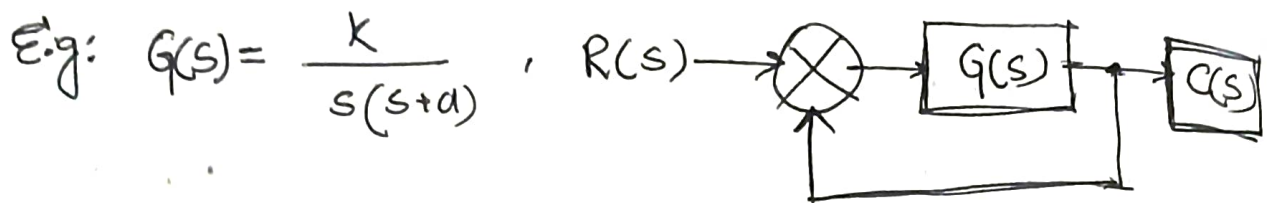
$$\therefore (z-1)^3 + 7(z-1)^2 + 25(z-1) + 39 = 0$$

$$\Rightarrow z^3 + 4z^2 + 14z + 20 = 0$$

The Routh array is -

z^3	1	14
z^2	4	20
z^1	9	0
z^0	5	

* Root locus:-



Open-loop poles located at $0, -a$

$$\text{closed loop T.F} = \frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{k}{s^2 + as + k}$$

$\therefore$ Characteristic eqⁿ, $q(s) = s^2 + as + k = 0$

$$s_{1,2} = \frac{-a \pm \sqrt{a^2 - 4k}}{2}$$

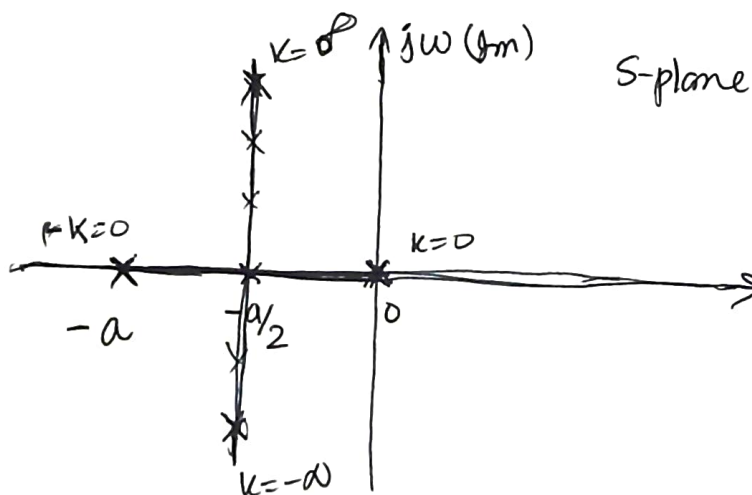
$$= -\frac{a}{2} \pm \sqrt{\frac{a^2}{4} - k}$$

Case 1:- when $0 \leq k < \frac{a^2}{4}$

1. $\frac{k=0}{k=0}$ $s_{1,2} = 0, -a \rightarrow$ Same as open loop.

2. $k = \frac{a^2}{4}$, $s_{1,2} = -\frac{a}{2}, -\frac{a}{2}$

3. $\frac{a^2}{4} < k < \infty$, $s_{1,2} \rightarrow$ Complex conjugate with real parts $-\frac{a}{2}$

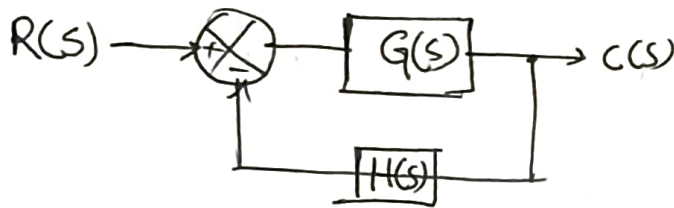


From Root locus Eqⁿ

$\rightarrow$ locus of the ~~is~~ characteristic Eqⁿ for different k values

$\rightarrow$ Stability of system for certain range of k .

* Construction Rules:-



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

∴ characteristic eqn: $q(s) = 1 + G(s)H(s) = 0$

$$\Rightarrow G(s)H(s) = -1 = \cos 0 + j \sin 0$$

$$|GH| = 1, \angle GH = (2q+1)180^\circ, q = 0, 1, 2, \dots$$

* Magnitude criterion, $|GH| = 1$, phase criterion, $\angle GH = (2q+1)180^\circ, q = 0, 1, 2, \dots$

→ The point will be on root locus, if both the criterion are satisfied, may be stable or unstable for certain 'K' values.

$$GH = \frac{K(s+z_1)(s+z_2)\dots}{(s+p_1)(s+p_2)\dots}$$

$$\text{Charac. Eqn} = 1 + GH = 0$$

$$\therefore 1 + \frac{K(s+z_1)(s+z_2)\dots}{(s+p_1)(s+p_2)\dots} = 0$$

$$\text{or, } 1 + \frac{K \prod_{i=1}^m (s+z_i)}{\prod_{j=1}^n (s+p_j)} = 0 \quad \angle \frac{s+a}{s+b} = \angle s+a - \angle s+b$$

$$\text{or, } \frac{K \prod_{i=1}^m |s+z_i|}{\prod_{j=1}^n |s+p_j|} = 1 \quad (\text{Magnitude criterion})$$

* Phase criterion:-

$$\sum_{i=1}^m \angle (s+z_i) - \sum_{j=1}^n \angle (s+p_j) = (2q+1)180^\circ, q = 0, 1, 2, \dots$$

* Construction:-

Rule 1:- The root locus is symmetrical about the real axis.

Rule 2:- As K varies from 0 to ∞ each branch of the root locus originates from an open loop pole with $K=0$ and terminates, either on an open loop zero or infinity: $K \rightarrow \infty$

The no. of branches terminating on infinity equals -
= no. of open loop poles - zeros

$$1 + \frac{K \prod_{i=1}^m (s + z_i)}{\prod_{j=1}^n (s + p_j)} = 0$$

$$\Rightarrow K \prod_{i=1}^m (s + z_i) + \prod_{j=1}^n (s + p_j) = 0$$

for, $K=0$, $s = -p_j$, $j=1, \dots, n \rightarrow$ Root locus starts from open loop poles

Also, $\prod_{i=1}^m (s + z_i) + \frac{1}{K} \prod_{j=1}^n (s + p_j) = 0$

$$K \rightarrow \infty \Rightarrow s = -z_i, i=1, \dots, m$$

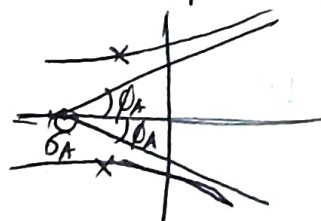
*** $\rightarrow$

lies on

3. A point on the real axis ~~is a~~ root locus, if the no. of open loop (poles + zeros) on the real axis to the right of this point is odd.

4. The $(n-m)$ root locus branches which tend to infinity do so along straight line asymptote whose angle are given by-

$$\phi_A = \frac{(2q+1)180^\circ}{n-m}; q=0, 1, 2, \dots, n-m-1$$



* eg:- $s^5 + 2s^4 + 8s^3 + 12s^2 + 20s + 16 = 0$

→ The Routh array is -

s^5	1	8	20	16
s^4	2 ₁	16 ₆	16 ₈	
s^3	2 ₁	16 ₆	16 ₈	
s^2	0	0		
s^1				
s^0				

Now, the auxiliary polynomial is -

$$A(s) = s^4 + 6s^2 + 8$$

$$\therefore \frac{dA(s)}{ds} = 4s^3 + 12s$$

s^3	1	8	20	16
s^2	1	6	8	
s^1	1	6	8	
s^0	1	12	3	
s^2	3	8		
s^1	$\frac{1}{3}$	0		
s^0	8			

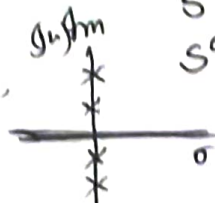
$$\frac{\frac{1}{3} \times 8}{\frac{1}{3}}$$

Now,

$$A(s) = s^4 + 6s^2 + 8 = 0$$

$$\Rightarrow s = \pm j\sqrt{2} \text{ and,}$$

$$s = \pm j2$$

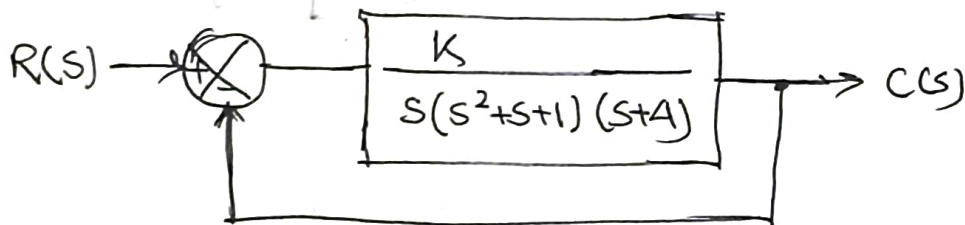


Note:- Because of row of zero in the array, the Routh's test breaks down, this situation is overcome by replacing the row of zero by a row of co-efficients of the polynomial generated by taking the first derivative of the auxiliary polynomial. Since, there is no sign changes, we will conclude that no root of characteristic equation ~~has~~ have real part.

$\therefore s = \pm j\sqrt{2}$ and $s = \pm j2$, these two pairs of roots are also the roots of the originally characteristic equation.
 $\therefore$ the system is limitedly (marginally) stable.

* Application of Routh Stability criterion:-

Note:- The Routh stability criteria is frequently used for the determination of stability condition of linear feedback control system. Consider the closed loop system, and let us determine the range of 'K' for which the system is stable.



$$\therefore \text{CLTF} = \frac{C(s)}{R(s)} = \frac{K}{s^4 + 5s^3 + 5s^2 + 4s + K}$$

The Routh array is -

s^4	1	5	K
s^3	5	4	
s^2	$21/5$	K	
s^1	$\frac{84/5 - 5K}{21/5}$		
s^0	K		

$\therefore$ For stability, $\frac{84}{5} - 5K > 0$ and $K > 0$

$\frac{84}{25} - K > 0$

$\therefore 0 < K < \frac{84}{25}$

when, $K = \frac{84}{25}$

$$A(s) = \frac{21}{5} s^2 + \frac{84}{25} = 0$$

$$\Rightarrow s = \pm j \frac{2}{\sqrt{5}} = j(\omega)$$

It causes sustained oscillation in the closed loop system and the oscillation frequency is $\frac{2}{\sqrt{5}}$ rad/sec.

5) The asymptotes cross the real axis at a point on an centroid determined by -

$$\sigma_A = \frac{\text{Sum of real parts of poles} - \text{Sum of real parts of zeros}}{n - m}$$

$n \Rightarrow$ no. of open loop poles

$m \Rightarrow$ no. of open loop zeros

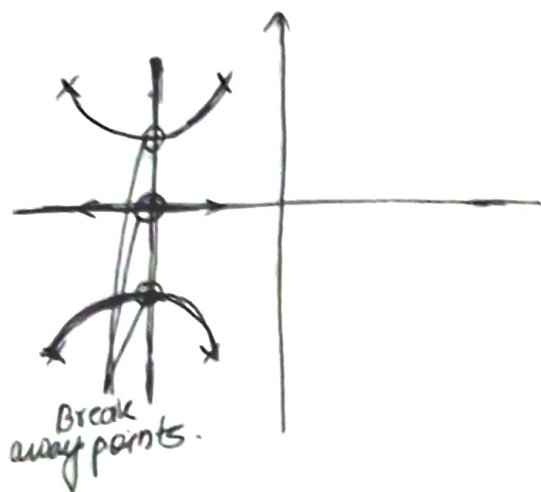
(6) The break away points... (multiple roots of the characteristic Eq. of the root locus are the solution of $\frac{dK}{ds} = 0$)

eg: $q(s) = s^3 + s^2 + s + K = 0$

$$\Rightarrow K = -s^3 - s^2 - s$$

$$\frac{dK}{ds} = 0 \rightarrow s = \dots$$

* Breakaway points are those points that satisfies magnitude & phase criteria.
(multiple points of root locus)



Note:- ① Root locus can have real as well as complex breakaway points

② Not all the roots of $\frac{dK}{ds} = 0$, correspond to the actual break-away point, the actual breakaway point are those at which the angle criteria is met.

Note:-

③ The root locus branches must approach or leave the break-away point on the real axis at an angle $\pm \frac{180^\circ}{r}$, where r is the no. of branches approaching or leaving the branch point

Rule 7:- The angle of departure from an open loop pole is given by

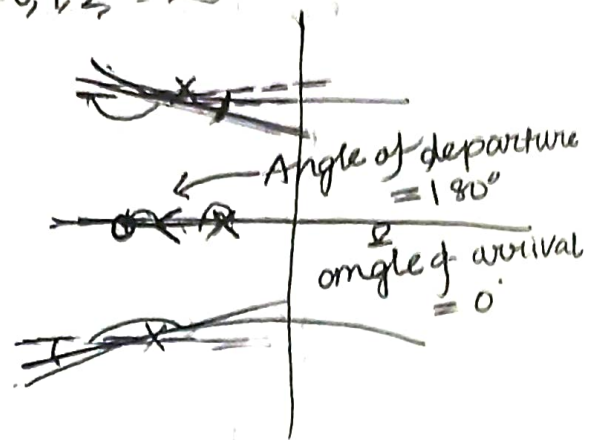
$$\phi_p = \pm 180^\circ (2q+1) + \phi; \quad q = 0, 1, 2, \dots$$

where ϕ is the net angle ~~consideration~~ contribution at this pole, of all other open loop poles & zeros.

Similarly, the angle of arrival at an open loop zero is -

$$\phi_z = \pm 180^\circ (2q+1) - \phi, \quad q = 0, 1, 2, \dots$$

Angle of departure for real open loop pole and angle of arrival for real open loop zero is always 0° or 180°

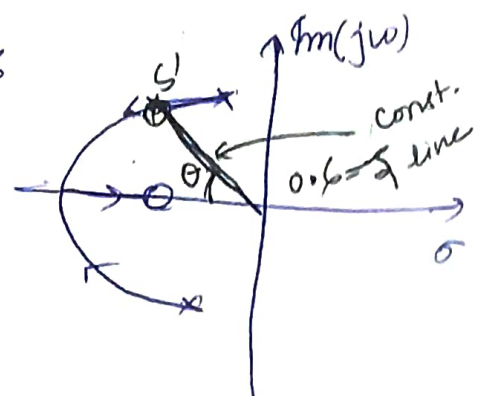


Rule 8: The intersection of the root-locus branches with the imaginary axis can be determined by use of the Routh criterion

Rule 9:- The open-loop gain in pole-zero form at any point s_0 on the root locus is given by

$$K = \frac{\prod_{j=1}^n |s_0 + p_j|}{\prod_{i=1}^m |s_0 + z_i|} = \frac{\text{Product of phasor lengths from } s_0 \text{ to open loop poles}}{\text{Product of phasor lengths from } s_0 \text{ to open loop zeros.}}$$

e.g.: Find the roots corresponding to $\zeta = 0.6$

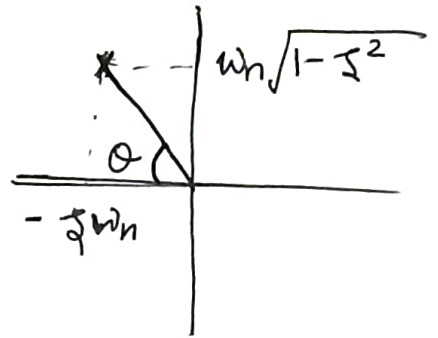


Soln: $\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

$\therefore$ characteristic eqn

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\therefore s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$



$$\cos \theta = \frac{+\zeta\omega_n}{\sqrt{\zeta^2\omega_n^2 + \omega_n^2 - \omega_n^2\zeta^2}}$$

$$\cos \theta = \frac{+\zeta\omega_n}{\omega_n}$$

$$\Rightarrow \theta = \cos^{-1}(\zeta)$$

eg: A feedback control system has CLTF -

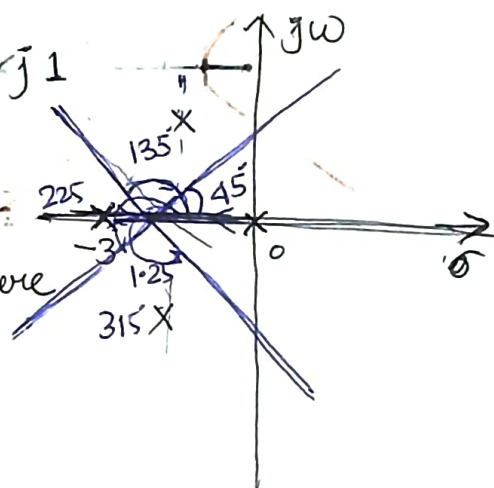
$$G(s)H(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

Find the root locus as K vary from zero to infinity.

Soln:- ~~characteristic eqn:-~~

The open loop poles are: 0, -3, -1 ± j1

All four branches ~~at~~, will go to infinity along straight line asymptotes, since there are no open loop zeros.



Asymptote angles are given by -

$$\phi_A = \frac{(2q+1)180^\circ}{n-m}, q=0,1,2,3$$

$$= \frac{(2q+1)180^\circ}{4-0}, q=0,1,2,3$$

$$= 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

Centroid:-

$$\sigma_A = \frac{(0-3-1-j-1)}{4-0} = -1.25$$

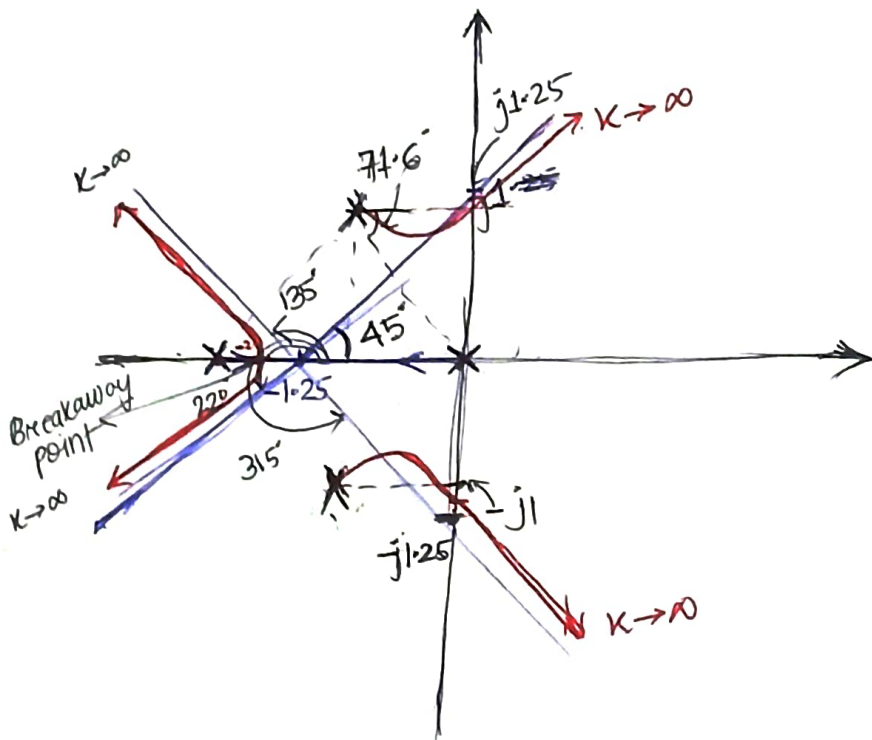
→ The real axis segment 0 to -3 is a part of the root locus.

Characteristic eqn: -

$$1 + \frac{K}{s(s+3)(s^2+2s+2)} = 0 \quad \text{Lam}$$

$$\therefore K = -s(s+3)(s^2+2s+2)$$

$$\therefore \frac{dK}{ds} = 0 \Rightarrow s = -2.3, -0.725 \pm j0.365$$



The angle of departure from ~~open loop~~ upper complex pole $(-1 \pm j1)$

$$\begin{aligned} \phi_p &\equiv \pm 180^\circ (2q+1) + \phi \\ &\equiv \pm 180^\circ + (-135^\circ - 90^\circ - 26.6^\circ) \\ &= -71.6^\circ \end{aligned}$$

Characteristic eqn:

$$1 + \frac{K}{s(s+3)(s^2+2s+2)} = 0$$

$$\therefore s^4 + 5s^3 + 8s^2 + 6s + K = 0$$

The Routh array is -

$$\begin{array}{cccc}
 s^4 & 1 & 8 & K \\
 s^3 & 5 & 6 & \\
 s^2 & \frac{34}{5} & K & \\
 s^1 & \frac{204}{5} - 5K & 0 & \\
 s^0 & \frac{34}{5} & & \\
 s^0 & K & &
 \end{array}$$

When,

$$\frac{\frac{204}{5} - 5K}{\frac{34}{5}} = 0$$

$$\Rightarrow K = \frac{204}{25}$$

$\therefore$ The auxiliary eqn is -

$$\frac{34}{5} s^2 + \frac{204}{25} = 0$$

$$\therefore s = \pm j 1.1$$

Q. Find the roots corresponding to $\xi = 0.7 \rightarrow$ constant $\xi = 0.7$ line with angle $\cos^{-1}(\xi)$

Q. Find 'K' at breakaway point.

Characteristic eqn.

$$1 + \frac{K}{s(s+3)(s^2+2s+2)} = 0$$

$$\Rightarrow -K = \left| s \right| \left| s+3 \right| \left| s^2+2s+2 \right| \Big|_{s=-2.3}$$

$$\Rightarrow K = 4.54$$